Difference Approximation Network With Semantic Alignment for Remote Sensing Change Captioning

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***Abstract*—Remote sensing change captioning (RSCC) plays an important role in advancing remote sensing spatiotemporal understanding with vision-language generation. However, due to the temporal evolution of geospatial patterns, bitemporal images are susceptible to the influence of lighting conditions and seasonal effects. Existing methods fail to effectively utilize the cross-sample difference information and perform explicit multimodal alignment of key changes, which often results in inaccurate descriptions of changes in key objects and their relationships. To address these issues, the article proposes a novel difference approximation network with semantic alignment (DANSA) for RSCC. Specifically, a cross-temporal feature transform (CFT) is introduced to interact with feature representations extracted from the contrastive language-image pretraining (CLIP) model. Then, to learn a reliable difference representation, a difference memory approximation (DMA) is designed to enhance the difference representation of the current scene by activating difference information from training samples. Furthermore, considering that pretrained language embedding models can serve as a powerful supervisory signal, a change semantic alignment (CSA) is constructed to achieve fine-grained alignment between visual and textual features. In this way, DANSA generates comprehensive scene change captions that accurately depict objects and relationships across time. Extensive experiments and analyses conducted on three benchmark datasets demonstrate that DANSA achieves competitive performance with fewer training parameters. The code will be available at https://github.com/zcsisiyao/DANSA**

***Index Terms*—Change captioning, contrastive language-image pretraining, cross-modal alignment, remote sensing.**

I. Introduction

With the rapid accumulation of multitemporal remote sensing data and advances in multimodal technology, geospatial artificial intelligence (GeoAI) has become a core paradigm for automated Earth surface monitoring and dynamic analysis [1], [2], [3], [4]. Among GeoAI tasks, remote sensing change captioning (RSCC) [8], [9], [10], [11] stands out as a critical multimodal spatiotemporal interpretation technique. Different from single-image interpretation tasks, RSCC aims to automatically generate human-readable natural language descriptions for land cover changes between bitemporal images of the same region. It requires both accurate perception of spatial patterns (e.g., object categories and spatial layouts) and precise modeling of temporal dynamics (e.g., object emergence, disappearance, and expansion), thus providing intuitive semantic support for urban expansion monitoring, deforestation tracking, and postdisaster reconstruction assessment [9], [12], [14].

Compared with the remote sensing image captioning (RSIC) [15], [16], [17], [18], RSCC faces unique challenges rooted in its dual-temporal nature, as shown in Fig. 1. Due to the influence of temporal resolution on multitemporal imagery, it is susceptible to false variations caused by changes in lighting conditions and seasonal shifts [19], [20]. The main challenges for RSCC can be summarized in two aspects. First, due to the distinctive characteristics of bitemporal images, it is difficult to achieve effective abstraction and representation of change targets and interrelationships. Second, the lack of interaction and alignment between extracted change features of image

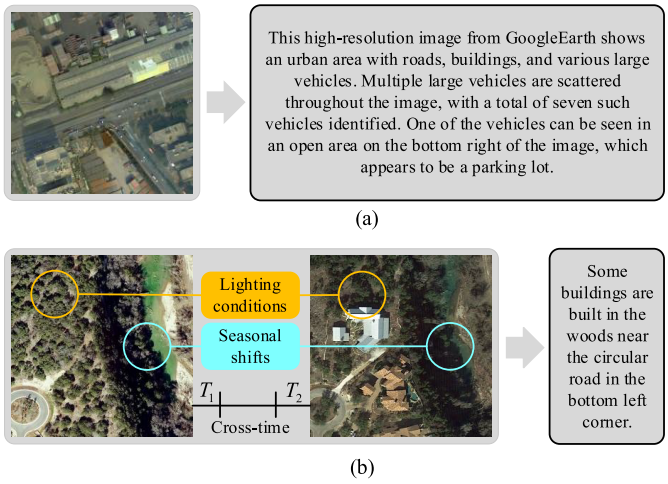


Fig. 1. Challenges of RSCC compared to RSIC tasks. Different colored circles correspond to different challenges: orange represents different lighting conditions, and blue represents different seasonal shifts. (a) Remote sensing image captioning. (b) Remote sensing change captioning.

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pairs and text features leads to semantic inaccuracies in the generated sentences.

Currently, mainstream RSCC methods [14], [21], [22] based on the encoding–decoding paradigm [23], [24], [25], [26] have been proposed to overcome these issues. These methods typically comprise three stages, including image feature extraction, difference feature representation, and change caption generation. For the feature extraction stage from bitemporal images, shared feature encoders [19], such as convolutional neural networks (CNNs) [27] or Vision Transformers (ViTs) [28], are employed as visual encoders to capture image features. To mitigate pseudo-changes in image pairs caused by temporal gaps, a shared feature encoder has been selected for extracting common representations from multitemporal images [29], [30]. Subsequently, to capture actual differences in images across time, a difference feature representation is constructed by integrating the extracted features from two time points and utilizing various attention mechanisms [20], [28], [31]. Finally, in the change caption generation stage, the image features before and after the change, along with the difference features, are jointly interacted with natural language embedding [32]. Deep autoregressive models, such as recurrent neural networks (RNNs) [33] or Transformers [34], are employed as language decoders to convert change features into natural language.

Despite continuous performance improvements, two critical limitations remain unsolved in current RSCC frameworks, which restrict caption quality in complex real-world scenarios. First, existing difference modeling relies heavily on a single current image pair to construct difference features, lacking global difference knowledge for feature calibration. Although shared encoders alleviate simple pseudo-changes to some extent, they fail to handle complex and diverse false variations in real remote sensing scenes [19], [31], [58]. Without aggregating change patterns from global training samples, the model is prone to misinterpreting seasonal or illumination changes as real land cover transitions, resulting in unreliable difference representations. Second, the cross-modal alignment between change features and language semantics remains implicit and insufficient. Most existing methods only use standard cross-attention to realize interaction between visual features and word embeddings [55], [57]. This implicit fusion lacks explicit constraints on the semantic correspondence between change targets and descriptive words, easily causing semantic deviations such as misidentified change types, mismatched object attributes, and inaccurate change intensity descriptions.

To alleviate these problems, this article proposes a novel difference approximation network with semantic alignment (DANSA) for RSCC, which simultaneously enhances difference feature robustness and realizes explicit cross-modal semantic alignment. Specifically, to reduce interference from false changes, a cross-temporal feature transform (CFT) is constructed to improve the feature representation extracted from the contrastive language-image pretraining (CLIP) model [35]. Considering that accessing historical training samples during the generation phase is regarded as a powerful source of information for enhancing description quality [36]. Building on the difference features of the current bitemporal images, a difference memory approximation (DMA) is designed to capture difference information extracted from global training samples. When describing the change information of the current image pairs, the DMA module fully considers the relevant difference information jointly observed from historical input samples. To better align the semantic features of change information, change semantic alignment (CSA) is constructed to integrate enhanced image features and difference features to match the semantics of change captions through optimal transport, which helps to bootstrap the model to acquire language-aligned features. Finally, complete and accurate change captions are generated by introducing a change captioning decoder with causal self-attention. In summary, the main contributions of this article are as follows.

1) A novel DANSA is proposed for the RSCC task. This network strengthens difference feature aggregation and cross-modal CSA, which bootstraps the change caption generation for remote sensing images.
2) To capture genuine changes in cross-temporal remote sensing images, CFT is designed to effectively interact with image features before and after changes. DMA is further constructed to enhance the aggregation between the current difference information and global difference information, thereby enabling better learning of difference representations in complex scenes.
3) CSA integrates difference features and image features, and leverages semantic information from language embedding models to guide the assignment of change features, thereby achieving fine-grained cross-modal semantic alignment.

The remainder of this article is structured as follows. Section II presents a review of related work. Section III elaborates on the overall framework of DANSA and provides a detailed explanation of each component within DANSA. To evaluate our proposed approach, experiments are designed in Section IV. Finally, this article concludes with a summary in Section V.

II. Related Work

In this section, we summarize the relevant works from two aspects: RSIC and RSCC.

A. Remote Sensing Image Captioning

In the field of GeoAI, RSIC focuses on describing the target objects within a single scene and their relational attributes. RSIC algorithms based on encoder–decoder architectures are employed [37], [38], [39], [40], and they are capable of adapting to differences in data sources, feature translations, and semantic representations. Qu et al. [42] adopt a CNN-RNN framework and demonstrated on the constructed UCM-captions and Sydney-captions datasets that the framework performs equally well on remote sensing images, even though it was originally designed for natural images. Due to extensive coverage and abundant information in remote sensing images, it is difficult to generate accurate semantic descriptions of crucial objects and their relationships. PKG-Transformer [43] enriches object and scene features using object detection and integrates these relationships into the

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Transformer encoder as prior knowledge. Zhang et al. [44] propose an interactive word network-enhanced Transformer for RSIC by jointly using CNN and ViT encoders. It obtains comprehensive features of multiscale targets and global relationships. In particular, to address the issue of limited training data, Kandala et al. [45] employ an auxiliary decoder for multilabel scene classification. This decoder leverages the conceptual similarity between the encoder and image captions, as well as its ability to highlight semantic categories, thereby assisting the encoder during training. Ye et al. [46] propose an approach that integrates two-stage training by fusing label attributes with image features through an attribute-guided decoder.

To better align image and text features, it becomes a compromise to directly apply image encoders that have been pretrained with vision and language alignment [41], [47], [48], [49]. Developments in vision-language pretraining have shown promising results in the field of vision tasks [50]. CLIP [35] provides a powerful pretrained model capable of learning shared representations of images and text. For example, Lin et al. [51] integrate CLIP's grid-based visual features with a random masking strategy in the RSIC task. Moreover, Meng et al. [52] utilize a hybrid image encoder composed of CLIP and CNN for richer image representation and further focus on different contextual information using a global grouping attention mechanism. Liu et al. [53] combine features extracted by CLIP's text encoder and image encoder, along with a graph attention network, to achieve semantic-spatial feature fusion and dynamic graph feature refinement. Ricci et al. [54] employ CLIP to filter or generate the single most harmonious caption from predictions produced by different captioning algorithms, thereby offering high flexibility for embedding diverse captioning algorithms with distinct characteristics and advantages. The CLIP-based RSIC technology holds strong potential for application in the field of RSCC.

B. Remote Sensing Change Captioning

Whereas RSIC deals with single-time image captioning, RSCC focuses on recognizing change objects and their relationships in same-area image pairs taken at distinct times [32]. This requires overcoming interference from illumination changes and seasonal shifts induced by the temporal gap. Change detection methods [63], [83] can to some extent overcome these pseudo-changes and identify actual alterations. However, pixel-level changes alone are unable to directly convey high-level semantics, such as the attributes of objects in the changed region and their interrelationships [58], which inevitably requires substantial manual interpretation. Chouaf et al. [59] first constructed a private change captioning dataset and proposed a model that generates captions by an RNN conditioned on the concatenated feature extracted by a shared pretrained VGG16. Hoxha et al. [60] design an encoding architecture that fuses image-level and feature-level information, combined with a decoding framework based on an RNN and support vector machines, to characterize meaningful differences between similar images. Subsequently, Liu et al. [61] introduce a dual-branch Transformer for change captioning and created the LEVIR-CC dataset to support model training and validation. The availability of remote sensing change description datasets plays a key role in advancing change description techniques. By analyzing representative examples drawn from natural images, synthetic image datasets, and remote sensing data, Chang and Ghamisi [62] emphasize the distinctive properties of remote sensing image pairs. They propose Chg2Cap, a method that uses an attention encoder together with a Transformer-based decoder to generate change descriptions for remote sensing images. Shi et al. [19] propose a multitask network jointly for change detection and RSCC, whose shared encoder benefits both tasks but whose difference representation remains confined to the current image pair. Gao et al. [63] combine SAM with limited data for change detection, demonstrating that foundation models can be effectively adapted with weak supervision. Li et al. [82] propose intertemporal interaction and symmetric difference learning for RSCC, yet their method relies solely on the current image pair without memory augmentation. Ning et al. [83] address cross-resolution change detection via frequency-domain unequal relationships. In addition to dominant change captioning methods based on CNN and Transformer technologies, probabilistic diffusion techniques have also been introduced into RSCC tasks [56]. Sun et al. [64] propose a paradigm shift toward data distribution learning via diffusion models, which is strengthened through frequency-domain noise filtering to offer a theoretically grounded and practically feasible solution for the multimodal RSCC.

Recently, CLIP-based methods have also been preliminarily explored in change captioning tasks [65]. Hang et al. [66] introduce predefined text prompts for remote sensing images, enhance the corresponding semantic features of images using CLIP, and jointly utilize features extracted by FastSAM [67] to describe changes in remote sensing images. This simple approach of leveraging category information retrieved via CLIP expands the semantic information of the target, but it lacks the alignment relationship between bitemporal image change information and the semantics of the change caption. Therefore, the potential for utilizing the CLIP model to explore change information alignment remains largely unexplored. To enhance the alignment of change information using CLIP, CFT is introduced to extract aligned bitemporal image features. These change visual features are then aligned with the frozen CLIP-extracted change text features using DMA-enhanced difference features, thereby achieving better change alignment and generating accurate change captions.

III. Methodology

The architecture of the DANSA is shown in Fig. 2. The DANSA consists of a CFT, a DMA, a CSA, and a change decoder. The DANSA mainly uses CFT to construct deep features across different time phases, enabling interactions between these features to enhance the model's ability to identify changing targets under varying imaging conditions. Subsequently, by constructing the difference representation as a query, DMA extracts the most relevant difference items in the memory via the attention-driven addressing operator. To better align with the semantics of change text, CSA is performed to align image features and difference features, which are finally

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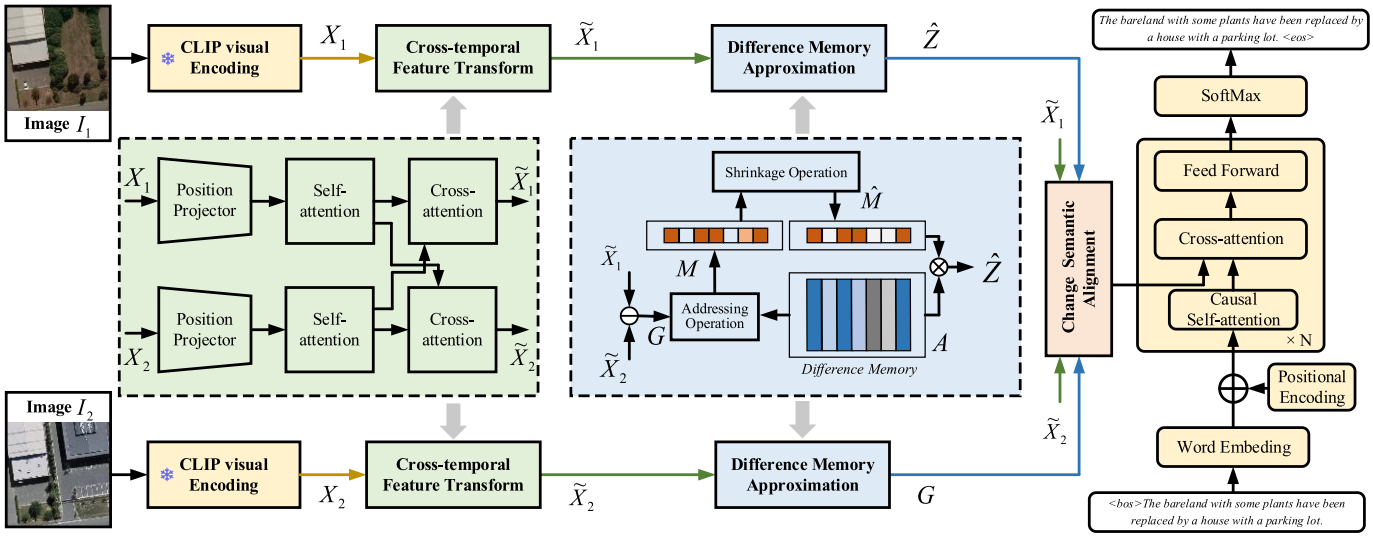


Fig. 2. Framework of the proposed DANSA. The CFT employs a position projector and MHA to interact bitemporal image features. The DMA stores cross-sample difference feature representations. The CSA enables fine-grained alignment of change textual features and change visual features. The change decoder with a Transformer generates captions with appropriate change objects and correlations.

delivered to the decoder for change caption generation. In this section, CFT, DMA, CSA, change decoder, and model learning are described in detail.

A. Cross-Temporal Feature Transform

Due to imaging differences between temporal phases, the encoding process using shared feature extractors is often susceptible to interference from false changes [64], and there is a lack of effective interaction between features from different temporal phases. To comprehensively capture targets and their interdependencies in dual-temporal remote sensing images, CFT employs CLIP-derived image features for dual-temporal feature acquisition, followed by a joint attention mechanism to facilitate feature interaction. The CLIP model has been pretrained with language alignment and can be well applied to vision-language tasks. Different attention forms enable effective interaction between temporal features. Furthermore, the integration of a position projector mechanism allows bitemporal features to more accurately perceive cross-temporal spatial relationships.

Formally, given two remote sensing images, I_1 and I_2, image features are extracted using a CNN visual encoder from CLIP. The feature extraction process is denoted as

$$X_i = CLIP_{visual}(I_i) \tag{1}$$

where $X_i \in \mathbb{R}^{C\times H\times W}, i = 1, 2$ denotes the bitemporal image features and C, H, and W indicate the number of channels, height, and width, respectively. $CLIP_{visual}$ corresponds to a frozen pretrained CLIP image encoder.

In fact, CFT discards the final pooling layer of the CLIP visual encoder and utilizes the visual feature X_i that preserves the original spatial features. Then, a sequence of operations comprising Convolution, ReLU, and Dropout is applied to the generated features to eliminate redundant information. The overall process is defined as

$$X_i' = \text{Dropout}(\text{Relu}(\text{Conv}(X_i))) \tag{2}$$

where bitemporal features $X_1'X_2'$ are obtained with the same channel dimension, where $X_i' \in \mathbb{R}^{N\times H\times W}$.

To better interact with language features and preserve spatial positional information, X_i' is reshaped as $X_i' \in \mathbb{R}^{D\times N}, D = HW$. Then, positional encodings [68] are introduced on the flattened feature set X_i'

$$\hat{X}_i = X_i' + \text{PE}(X_i') \tag{3}$$

where PE is positional encodings. $\hat{X}_1$ and $\hat{X}_2$ are transformed feature representation used for attention interaction.

Subsequently, $\hat{X}_1$ and $\hat{X}_2$ interact both within and across features through multihead attention (MHA) [34], thereby generating bitemporal aggregated semantic features $\tilde{X}_1$. Similarly, $\tilde{X}_2$ can also be obtained in the following ways. The intermediate process is as follows:

$$X^1 = \text{MHA}(\hat{X}_1, \hat{X}_1, \hat{X}_1) \tag{4}$$

$$\tilde{X}_1 = \text{MHA}(X^1, X^2, X^2) \tag{5}$$

where X^i is a feature obtained through self-attention. MHA relies on scaled dot-product attention and comprises h distinct attention heads, each of which is independent of the others and calculated in parallel, and concatenates the outputs of all heads

$$\text{MHA}(Q, K, V) = \text{Concat}_{i=1,\ldots,h}(\text{head}_i) \tag{6}$$

$$\text{head}_i = \text{Attn}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{7}$$

$$\text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}, \text{dim} = 1\right)\text{V} \tag{8}$$

where Attn is scaled dot-product attention and W_i^Q, W_i^K, and W_i^V are the learnable parameters.

B. Difference Memory Approximation

After obtaining the two cross-temporal feature representations, a straightforward approach is to perform a subtraction operation between them, which captures the difference

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between the two representations. However, the relevant differences in other bitemporal images have not been fully utilized, such as the evolution of the same natural features across different regions. To address this limitation, the DMA is proposed to specifically construct a learnable prototype memory bank in the difference feature space, which is used to approximate and calibrate the original difference representations. Different from general memory-based aggregation paradigms [5], [6], DMA models the distribution of typical change difference semantics via a set of compact, discriminative prototype vectors, and performs sparse content-addressable retrieval to aggregate globally representative difference information. In this way, it produces a more robust and semantically consistent global difference representation, which facilitates accurate change caption generation.

Given the enhanced bitemporal features $\tilde{X}_1$ and $\tilde{X}_2$, the difference feature G are first computed as

$$G = \tilde{X}_1 - \tilde{X}_2. \tag{9}$$

To exploit the difference information across samples, a difference memory is designed as a matrix $A \in \mathbb{R}^{E\times F}$ which contains E learnable prototype vectors of dimension F, where F has the same dimension as the difference feature G. Let the row vector a_j, for $j \in 1,\ldots,E$ denote the jth memory item. Given a difference query $g \in \mathbb{R}^F$ derived from G, the DMA produces a weighted combination of memory items via an attention weight vector $M \in \mathbb{R}^{1\times E}$

$$z = MA = \sum_{j=1}^{E} m_j a_j \tag{10}$$

where all elements of M are nonnegative and sum to one. The vector M is computed based on G, and the hyperparameter E controls the memory capacity.

The memory A is designed as a content-addressable structure [69] that dynamically updates during training to capture prototypical difference semantics. The addressing weights are computed via a SoftMax operation over cosine similarities

$$m_j = \text{SoftMax}\left(sim\left(g, a_j\right)\right) \tag{11}$$

where $sim(\cdot,\cdot)$ denotes cosine similarity.

By this design, DMA extracts the memory items most similar to g to approximate z. However, meaningful change semantics are inherently sparse within a scene, while unchanged background regions and spurious changes caused by inconsistent imaging conditions dominate the difference feature space. Activating excessive memory items with low relevance will introduce redundant background interference and ambiguous change semantics, which is detrimental to generating accurate change descriptions. To further enhance the discriminability of aggregated difference representations, we explicitly encourage sparse activation of memory prototypes. Although attention-based addressing inherently concentrates weights on a subset of items [70], the standard full SoftMax still produces a dense weight distribution with numerous tiny entries. This dense activation often incorporates irrelevant semantic differences into the final representation, such as changes in lighting, seasonal fluctuations in vegetation, and unchanging background patterns, thereby weakening the representation of changes in the actual target. To mitigate this, we introduce a hard shrinkage operation

$$\widehat{m}_j = \frac{\delta_j \cdot m_j}{\delta_j + \epsilon} \tag{12}$$

where $\delta_j = \max(m_j - \lambda, 0), m_j$ denotes the jth entry of the shrunk weight vector $\widehat{M}$ and ϵ is a small positive scalar. The threshold λ is set to $2/E$ empirically. After shrinkage, we renormalize weight vector as $\widehat{m}_i = \widehat{m}_i/\|\widehat{M}\|_1$. The final aggregated representation is given by $\hat{z} = \widehat{M}A$. This sparse addressing strategy prompts the model to activate fewer yet more relevant memory items, thereby generating a difference representation with enhanced global discriminability.

Owing to the limited memory capacity and the sparsity-inducing mechanism, only the most relevant difference semantics are retrieved during inference. Consequently, the proposed DMA module facilitates the generation of accurate change captions by leveraging compact and discriminative difference prototypes.

C. Change Semantic Alignment

To further align change features with text semantics, it is proposed to exploit the language embedding model [71] to bootstrap image features and difference features. Considering that the powerful language understanding capabilities of the CLIP text encoder have been thoroughly validated in [72] and [73], the text features extracted by this encoder are adopted as the foundation. For fine-grained CSA, directly computing the mean squared error between change features and linguistic embeddings not only introduces bias but also ignores the causality of the change object's attributes and dependencies in the order of sentences [74]. Inspired by [75], the alignment of change features is considered as an optimal transport problem, allowing each language embedding to match similar fine-grained change features.

To obtain the change features X^c of bitemporal image pairs, we first perform an elementwise addition on the difference features G and the approximate difference features $\hat{Z}$, similar to a residual connection

$$\tilde{\mathrm{G}} = \text{Relu}\left(\mathrm{G} + \hat{\mathrm{Z}}\right). \tag{13}$$

Then, we perform a linear transformation on the image features $\tilde{X}_1$ and $\tilde{X}_2$, as well as the enhanced difference feature $\tilde{G}$, to obtain the change feature X^c

$$X^c = W\left[\tilde{X}_1, \tilde{X}_2, \tilde{\mathrm{G}}\right] + b \tag{14}$$

where $W \in \mathbb{R}^{N\times C}$ and $b \in \mathbb{R}^N$ are the parameters to be learned. For the optimal transport involving changes in visual features and change language embeddings, suppose there are O suppliers and Q demanders, where the ith supplier has u_i units of goods, and the jth demander requires υ_j units of goods. Let c_{ij} be the cost of transporting one unit of goods from the ith supplier to the jth demander. The optimal transport problem aims to determine a transport plan, denoted as $\Omega = \{\omega_{ij} | i = 1,2,\ldots,M, j = 1,2,\ldots,N\}$, that minimizes the total transport cost C while ensuring that all goods are

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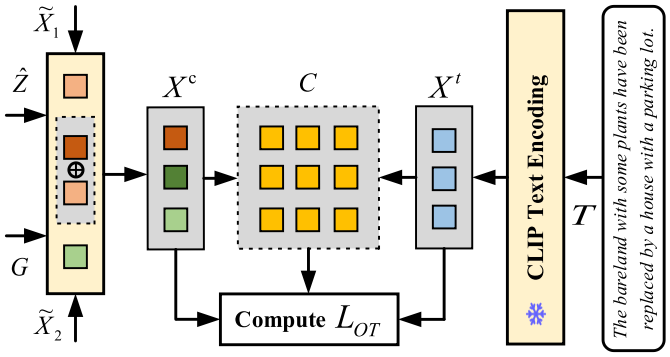


Fig. 3. Illustration of the CSA module, which uses the CLIP text encoder for the matching of change visual tokens to change semantic tokens.

delivered from the suppliers to the demanders. The process to compute OT is formulated as

$$\Omega = \mathrm{OT}\left(C, X^t, X^c\right) \tag{15}$$

where X^t represents the textual token, which is encoded using the CLIP text encoder.

The workflow for aligning semantic features is described in Fig. 3, considering N_t change textual tokens and N_c change visual tokens of the change semantic pair. Each change textual token is treated as a supplier, and each change visual token as a demander for assignment. The similarity between each value unit in the textual token and any value unit in the visual token denotes the transportation cost c_{ij}. This approach is extended to matrix-level mean squared error calculation to achieve GPU acceleration. Accordingly, the transportation cost from the ith language embedding to the jth change feature can be expressed as

$$c_{ij} = \left\| x_i^t - x_j^c \right\|_2^2 \tag{16}$$

where textual token set $X^t = [x_1^t, \ldots, x_{N_t}^t]$ and visual token set $X^c = [x_1^c, \ldots, x_{N_c}^c]$. The textual tokens closer in Euclidean distance to the visual tokens have higher similarity, tending to lower costs. A fast optimization algorithm, Sinkhorn–Knopp [76] is employed to calculate the transportation plan Ω in (15). By iteratively updating the strategy matrix between the textual token set and the visual token set, the scaled matrix Ω becomes the optimal transportation matrix that satisfies the marginal constraints. According to the solved $\Omega = \{\omega_{ij}\}$, the optimal transport loss allocates similar multiple language embeddings for jth change feature as assignment target, defined as follows:

$$L_{OT}\left(X^t, X^c\right) = \sum_{i=1}^{N_t} \sum_{j=1}^{N_c} \left\| x_i^t - x_j^c \right\|_2^2 \omega_{ij}. \tag{17}$$

By designing the above loss function, the model is able to capture changes in objects and their dependencies during cross-modal alignment. It should be noted that the proposed CSA module acts only as an auxiliary semantic alignment constraint during training. During inference, the model generates change captions solely based on change features, without relying on any text input or CLIP text embeddings. In fact, the CSA module performs optimal transport alignment at the individual token level, matching atomic semantic units such as land cover categories, change actions, and attribute relations. Therefore, it can achieve fine-grained alignment than traditional sentence-level alignment.

D. Change Captioning Decoder

The change decoder consists of a vanilla Transformer [34] equipped with causal self-attention, stacked across N identical layers. Each Transformer layer comprises a causal self-attention layer, a cross-attention layer, and a feed-forward network (FFN) layer. Residual connectivity and layer normalization are employed around each sublayer. Given the generated caption denoted as T, get the feature vector P embedded by the word embedding method and containing positional encoding, which is represented as follows:

$$P = \mathrm{Embedding}\,(T) + \mathrm{PE}\,(T) \tag{18}$$

where PE is positional encoding and Embedding is learning word embedding.

The input of each decoding layer is composed of the output from the preceding decoding layer and the change features produced by CSA, which are incorporated into the decoder via a cross-attention layer. Subsequently, the output F_d of the final decoding layer is taken as the decoder's output. Eventually, F_d is input to a fully connected layer followed by a SoftMax operation to compute the probability distribution over the vocabulary of candidate words.

E. Training and Objectives

Following a standard practice in change captioning [33], the DANSA is optimized to minimize a word-level cross-entropy loss. Specifically, given a dataset of paired images and captions $\{I_1^i, I_2^i, T^i\}_{i=1}^N$, our goal is to learn the generation of a meaningful change caption for an unseen input image pair. These captions can be regarded as a token sequence $T^i = [\mathrm{t}_1^i, \ldots, \mathrm{t}_l^i]$, in which the tokens are padded to the maximum length j. Accordingly, our training objective can be defined as follows:

$$L_c(\theta) = -\sum_{i=1}^{N} \log p_\theta\left(\mathrm{t}_1^i, \ldots, \mathrm{t}_j^i \middle| \left(I_1^i, I_2^i\right)\right) \tag{19}$$

where θ denotes the variable parameter of the model. L_{OT} and L_c are fixed with the same weights for model training. The DANSA is trained using parallel computation for enhanced learning efficiency.

IV. Experimental Study

In this section, extensive experiments on the RSCC dataset are performed to demonstrate the effectiveness of DANSA. First, the dataset, evaluation metrics, and experiment settings are introduced. Then, a comparison and ablation study are conducted and analyzed. We also provide some visual results. Finally, we conduct an in-depth quantitative analysis and discuss the limitations and future work.

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A. Datasets

In this study, to fully demonstrate the effectiveness of the DANSA, we conduct experiments on three public datasets, including the LEVIR-CC, WHU-CDC, and SECOND-CC datasets.

1) *LEVIR-CC [61]:* The LEVIR-CC dataset consists of 10 077 bitemporal remote sensing image pairs, with a spatial resolution of 0.5 m/pixel and a spatial size of 256×256 pixels. These image pairs are collected from 20 different regions in Texas, USA, and cover diverse urban and suburban construction processes across different time periods. Each pair of images is manually labeled with five descriptive sentences that characterize the observed changes, yielding a total of 50 385 change captions. This dataset supports the research and assessment of remote sensing change caption generation models. In this dataset, the maximum length of a caption sentence is 39 words, while the average length is eight words.
2) *WHU-CDC [19]:* The WHU-CDC dataset comprises 7434 pairs of images, each with dimensions of 256×256 pixels. The dataset considers the scenes depicted in bitemporal images, identifies the nature of the changes, and considers attributes and relative positions of the changes in relation to other objects. These bitemporal images were captured in 2012 and 2016 for detecting the earthquake reconstruction of buildings in Christchurch, New Zealand. The images have a resolution of 0.075 m. In total, the dataset contains 37 170 sentences. The maximum sentence length is 24 words, with an average length of nine words.
3) *SECOND-CC [55]:* The SECOND-CC dataset contains remote sensing images that have a size of 256×256 and provide resolutions ranging from 0.5 to 3 m per pixel. The image pairs in SECOND-CC capture the aerial views of three Chinese cities of Hangzhou, Chengdu, and Shanghai, each with a variety of terrain changes and distractors. Notably, the images are designed to reflect real-world conditions, including challenges such as blur, intensity and contrast differences, viewpoint variations, and misregistration of images. Overall, the dataset consists of 10 855 image pairs, containing 54 275 change descriptions.

B. Evaluation Metrics

Referring to existing research on RSCC, this study utilizes five quantitative evaluation metrics to measure model performance: BLEU, ROUGE, METEOR, CIDEr, and S_m. These metrics were calculated with the use of cocoapi [7]. The higher the score of each metric, the better the quality of the generated change captions.

1) *BLEU [77]:* Bilingual evaluation understudy is designed to assess the similarity between generated text and reference text. It places greater emphasis on precision, computing the score by determining the proportion of 1–4 g that exist in both the generated text and the reference text, followed by integrating exact and partial matching results.
2) *METEOR [78]:* Metric for evaluation of translation with explicit ordering integrates both exact and approximate matching criteria, such as synonyms and word stems. It leverages external dictionaries to assist in identifying synonyms and stems, thereby enabling a more comprehensive evaluation of the quality of the generated text.
3) *ROUGE-L [79]:* Recall-oriented understudy for gisting evaluation is an evaluation metric focused on measuring the recall rate between generated text and reference text. It assesses the similarity between the two types of text by analyzing the length of their longest common subsequence.
4) *CIDEr [80]:* Consensus-based image description evaluation emphasizes the diversity and coherence of generated captions. It derives scores by comparing word frequency distributions between the generated text and multiple reference texts, with a particular focus on ensuring consistency across all reference texts.
5) S_m *[43]:* This metric is computed as the arithmetic average of the BLEU-4, METEOR, ROUGE-L, and CIDEr scores, which allows for a comprehensive assessment of the quality of the generated captions.

C. Implementation Details

DANSA uses the CLIP pretrained [35] and frozen ResNet-101 network as an image feature extractor. The specific scale of features chosen is from the output of the conv4_x layers. For the three datasets, the number of memory banks E is chosen as 1000, with an embedding dimension of 512. For the optimal transport formulation, we adopt uniform marginal distributions to assign equal semantic weight to each token, which strictly satisfies the mass conservation constraint of balanced optimal transport. The entropy regularization coefficient is set to 10^{-2} to preserve fine-grained alignment discriminability. The Sinkhorn–Knopp algorithm runs with a maximum of 100 iterations and an early stopping threshold of 10^{-5} for marginal error. Before constructing the cost matrix, we exclude all special tokens and padding tokens from the CLIP text encoder outputs according to the attention mask, and retain semantically meaningful word tokens for OT alignment to avoid introducing irrelevant noise. The captioning decoder employs a vanilla Transformer structure, which consists of three cycles and eight parallel attention heads within the MHA mechanism. For the text generation process, we append a start token <bos> and an end token <eos>, respectively, before and after each ground truth sequence. The word embedding dimension is configured as 512. We set the batch size to 50 and optimized the model parameters using the Adam optimizer. The hidden dimension of each FFN layer is set to 2048. In addition, we defined "patience" to monitor the model's performance on the validation set and to decide whether the model needed to be improved.

During the training process, we adopt the learning rate scheduling strategy [43], which incorporates a warm-up operation comprising 3000 iterations. We keep the model with the highest BLEU-4 score on the validation set as an initialization

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model for subsequent training phases or inference. When the patience value reaches 5, the training focus shifts to BLEU-4 optimization [43], and the learning rate is fixed at 5×10^{-6}. In both the optimization and decoding phases, a beam size of 5 is adopted to generate candidate change sequences. All experiments are conducted on an Ubuntu system equipped with an NVIDIA GeForce RTX 5090 graphics card with 32 GB of memory, which facilitates the acceleration of model training.

D. Compared Methods

To verify the superiority of the DANSA, this work compares 14 representative state-of-the-art methods, which are listed as follows.

1) DUDA [33] acquires the change representation by subtracting the features extracted by temporal CNNs. A dual spatial attention mechanism is incorporated, which is translated into natural language via a basic RNN network.
2) MCCFormers-S [27] is a Transformer-based change captioning method that utilizes ResNet101 to extract bitemporal features and directly feeds these features into a Transformer for caption generation.
3) RSICCformer [61] is based on the Transformer architecture. This method extracts change features from images using a dual-branch Transformer and performs translation through a Transformer decoder.
4) ICT-Net [20] leverages difference information from two temporal sources to guide their interaction, thereby extracting and integrating the most critical change information.
5) PSNet [28] is a Transformer-based model that employs a multilayer progressive difference-aware stacking architecture to extract change information. In addition, a scale-aware enhancement is designed and integrated with the decoder layer to enable progressive utilization of multiscale features.
6) Prompt-CC [30] puts forward a multiprompt learning strategy, which is designed to effectively tap the potential of the large language model (LLM) and guide the LLM to grasp the decoupling result.
7) Chg2Cap [62] introduces an attention encoder that captures the change information between bitemporal images using a hierarchical self-attention and a change residual module. The change caption is then generated through a Transformer decoder.
8) Sparse Focus [31] is a Transformer-based method. To reduce the computational complexity introduced by the Transformer, SFT modifies the self-attention mechanism in the encoding part to a sparse focal attention mechanism and then generates the final sentence through the decoder.
9) SEN [81] adopts a model pretraining approach, designing a CNN encoder tailored for bitemporal image inputs. It is pretrained on the large multitemporal remote sensing dataset SSL4EO-S12, thereby improving the model's performance in processing bitemporal remote sensing images.
10) CTMTNet [19] employs a CNN-based weight-sharing twin network as its feature extraction backbone and introduces a novel feature enhancement fusion module. During training, an improved multitask loss function is utilized for supervised learning to achieve task balance.
11) Mask Approx Net [64] employs a diffusion model, which consists of a multiscale change detection module and a frequency-guided complex filtering module that mitigates high-frequency noise during the entire diffusion process.
12) MModelCC [55] integrates visual features through cross-modal cross-attention and single-modal difference cross-attention, and additionally proposes a multimodal gateway cross-attention mechanism to facilitate interaction between change features and text features.
13) KGBDNet [21] enhances the feature representation of temporal changes through a retrieval-based strategy, and further employs Mamba in combination with learnable prompts for temporal–spatial feature fusion.
14) D3_Net [57] performs operations on image features in both the spatial and frequency domains, and then generates change captions by designing a multiscale dual-domain fusion module.

E. Comparison With Existing Methods

For fair comparison with existing methods, the preprocessing operations for these datasets and the division into training, validation, and test sets are consistent with the original literature [19], [55], [61]. Furthermore, the training settings in the reproduced experiments are consistent with the configurations specified in the original paper.

Tables I–III present quantitative results on the LEVIR-CC, WHU-CDC, and SECOND-CC datasets. By comparing with state-of-the-art methods, it is evident that DANSA demonstrates superior performance across multiple metrics on both datasets. Methods based on the CNN-Transformer paradigm achieved competitive performance in the change description task. Specifically, on the LEVIR-CC dataset, the proposed method achieved competitive performance in terms of the BLEU-3, BLEU-4, METEOR, and S_m metrics. Based on the CNN-Transformer method, Chg2Cap achieves impressive results on both BLEU-1 and BLEU-2 metrics by enhancing and localizing image features before and after changes through the introduction of hierarchical self-attention and masked convolutional blocks. However, it also has the largest number of training parameters, as shown in Fig. 4(a). DANSA employs frozen encoders for feature representation, which to some extent alleviates the training parameters. It also performs well across most metrics. Compared to diffusion-based methods such as Mask Approx Net, the proposed method also outperforms them on most metrics.

In the experimental results of the WHU-CDC dataset presented in Table II, the proposed method achieved favorable results across all metrics except BLEU-1. Unlike LEVIR-CC, the WHU-CDC is smaller in scale and higher in resolution, making it more challenging to describe changes. Therefore, methods with stronger generalization capabilities are required. As can be seen from Fig. 4(b), the performance of the S_m

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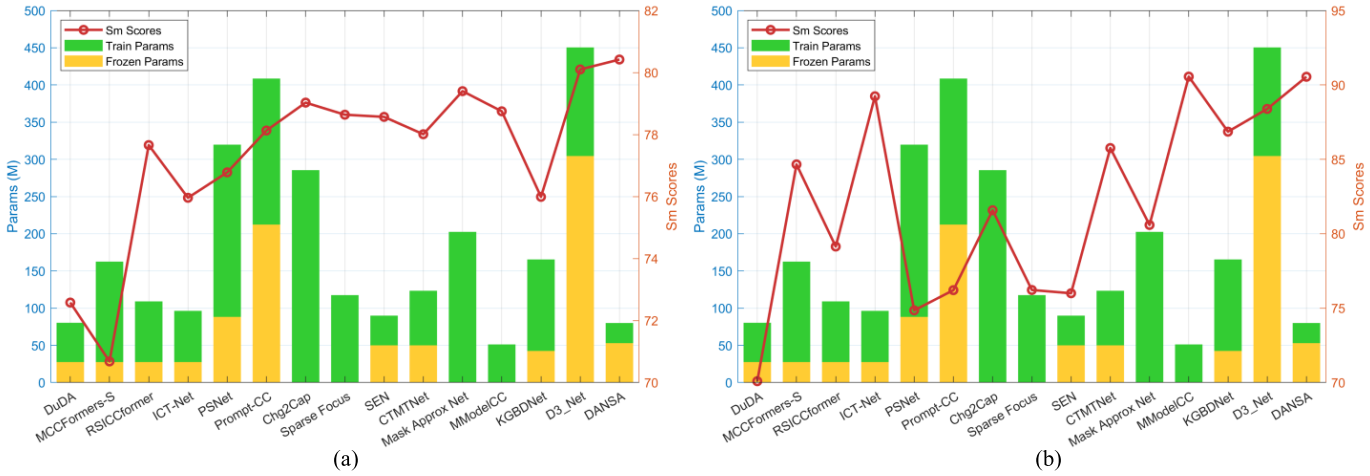


Fig. 4. Comparison of frozen parameters, trainable parameters, and metric S_m across different methods on (a) LEVIR-CC dataset and (b) WHU-CDC dataset.

TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LEVIR-CC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
DUDA	81.44	81.44	64.24	57.79	37.15	71.04	124.32	72.58
MCCFormers-S	79.90	70.26	62.68	56.68	36.17	69.46	120.39	70.68
RSICCformer	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.66
ICT-Net	84.45	75.69	67.94	61.54	38.83	73.25	130.23	75.96
PSNet	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
Prompt-CC	83.66	75.73	69.10	63.54	38.82	73.72	136.44	78.13
Chg2Cap	**86.14**	**78.08**	70.66	64.39	40.03	75.12	136.61	79.03
Sparse Focus	84.56	75.87	68.64	62.87	39.93	74.69	137.05	78.64
SEN	85.10	77.05	70.01	64.09	39.59	74.57	136.02	78.57
CTMTNet	85.16	77.24	70.13	64.16	39.34	74.03	134.52	78.01
Mask Approx Net	85.90	77.12	70.72	64.32	39.91	75.67	137.71	79.40
MModelCC	84.95	76.29	69.08	63.16	39.93	74.70	137.20	78.75
KGBDNet	83.57	75.23	68.26	62.78	38.39	72.80	129.99	75.99
D3_Net	85.18	77.04	69.92	63.86	39.14	**75.79**	**141.60**	80.10
DANSA	85.42	77.54	**70.96**	**65.72**	**40.76**	75.56	139.63	**80.42**

TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE WHU-CDC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
DUDA	79.04	69.53	61.57	55.64	34.29	68.98	121.45	70.09
MCCFormers-S	82.14	76.29	71.08	66.51	43.50	79.76	148.88	84.66
RSICCformer	80.05	74.24	69.61	66.54	42.65	73.91	133.44	79.14
ICT-Net	**86.34**	81.10	77.03	74.28	47.31	80.93	154.43	89.24
PSNet	81.26	73.25	65.78	60.23	36.97	71.60	130.52	74.85
Prompt-CC	81.12	73.96	67.22	61.45	36.99	71.88	134.50	76.21
Chg2Cap	78.93	72.64	67.20	62.71	41.46	77.95	144.18	81.58
Sparse Focus	81.17	72.90	66.06	60.27	37.34	72.63	134.64	76.22
SEN	80.60	74.64	67.69	61.97	36.76	71.70	133.57	76.00
CTMTNet	83.56	77.66	72.76	69.00	45.39	79.23	149.40	85.76
Mask Approx Net	81.34	75.68	71.16	67.73	43.89	75.41	135.31	80.59
MModelCC	85.89	80.79	76.65	73.53	**47.95**	**81.85**	157.70	90.26
KGBDNet	84.30	79.06	75.28	72.69	46.56	79.17	149.02	86.86
D3_Net	84.36	79.13	74.21	69.72	45.18	81.56	157.08	88.39
DANSA	86.27	**81.53**	**77.85**	**75.06**	47.70	81.21	**158.21**	**90.55**

metric exhibits greater variability across different methods on the WHU-CDC. The DANSA utilizes CLIP, pretrained on large-scale image–text datasets, as its backbone. This enables the extraction of image features aligned with textual representations, making them more suitable for the task of change captioning. In the experimental results for the SECOND-CC dataset presented in Table III and Fig. 5, the proposed method achieved leading results on five evaluation metrics. Since the SECOND-CC dataset includes a wider variety of scenarios, such as seasonal and lighting variations, it is more challenging.

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TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SECOND-CC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
RSICCformer	61.03	45.90	34.96	27.09	23.80	52.93	77.33	45.29
ICT-Net	62.45	46.86	36.27	29.06	23.88	49.44	64.94	41.83
PSNet	62.34	45.73	34.36	26.38	23.84	51.97	73.95	44.04
Prompt-CC	59.01	43.40	34.10	27.89	22.19	48.62	62.20	40.23
Chg2Cap	63.24	47.05	36.26	28.68	24.80	53.31	78.43	46.30
Sparse Focus	63.69	48.42	37.69	29.93	25.10	53.73	**81.49**	47.56
SEN	62.08	48.43	39.32	32.92	24.80	52.10	74.26	46.02
MModelCC	64.11	48.27	36.46	27.89	25.02	54.18	78.14	46.31
KGBDNet	64.15	48.73	38.60	31.66	25.18	52.68	73.78	45.83
D3_Net	62.94	49.87	40.66	34.26	**25.34**	**54.45**	78.02	48.02
DANSA	**65.09**	**51.11**	**41.47**	**34.79**	24.42	54.19	80.10	**48.38**

TABLE IV
RESULT OF ABLATION STUDY ON THE LEVIR-CC. ALL THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
Base	80.69	71.56	63.02	54.34	36.93	74.10	**149.91**	78.82
Base+CFT	81.35	72.65	64.63	56.64	37.67	75.04	149.81	79.79
Base+CFT+DMA	82.35	73.51	65.45	57.74	38.44	75.36	146.17	79.43
DANSA	**85.42**	**77.54**	**70.96**	**65.72**	**40.76**	**75.56**	139.63	**80.42**

TABLE V
RESULT OF ABLATION STUDY ON THE WHU-CDC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
Base	81.46	75.22	69.93	65.17	45.09	80.41	155.77	86.61
Base+CFT	83.22	78.08	74.39	71.75	45.71	77.86	147.65	85.74
Base+CFT+DMA	84.64	79.36	74.87	71.09	46.17	80.31	157.08	88.66
DANSA	**86.27**	**81.53**	**77.85**	**75.06**	**47.70**	**81.21**	**158.21**	**90.55**

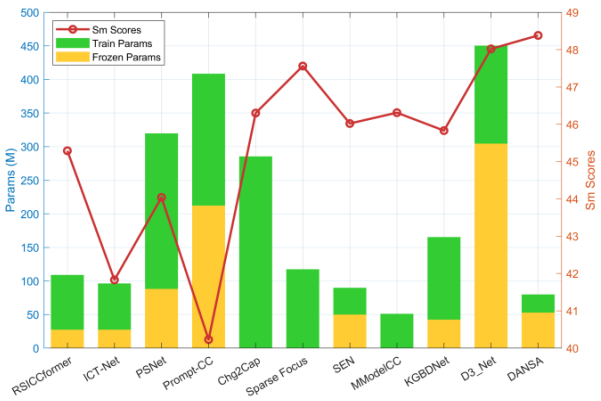


Fig. 5. Comparison of frozen parameters, trainable parameters, and metric S_m across different methods on SECOND-CC dataset.

Compared to the comparison methods, DANSA still achieves good results.

F. Ablation Study

The impact of the different components that comprise the DANSA on its overall performance is assessed through an ablation study. Specifically, the proposed DANSA employs only the CLIP-RN101 and the captioning decoder as the baseline model. All ablation variants share the exact same vanilla Transformer decoder. Details of the ablation analysis are given below.

1) *Effects of Each Component:* The proposed modules are added one by one to build the ablation model. These models are as follows: Base+CFT is baseline+CFT, which aims to validate the effectiveness of the CFT. The Base+CFT+DMA is baseline+CFT+DMA for the purpose of verifying the importance of supplementing the importance of change targets and their dependencies by utilizing the difference information from global samples within DMA. DANSA is a complete model that includes all components. The results are summarized in Tables IV–VI for the complete set of evaluation criteria used to evaluate the performance of the three change captioning datasets. For the effectiveness of the CFT module, a decomposition has been performed, and a comparison is performed through Base and Base+CFT, which shows that the introduction of the CFT module has improved in all change captioning datasets. By comparing Base and Base+CFT, it is found that interacting bitemporal features can generate more discriminative image representations, which alleviates, to some extent, the stylistic differences arising from different imaging times in bitemporal images. In addition, by comparing Base+CFT and Base+CFT+DMA, it is shown that global difference information is also important for generating accurate captions, with Base+CFT+DMA achieving performance improvements across most metrics. Ultimately, by comparing the metric results of Base+CFT+DMA and

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TABLE VI
RESULT OF ABLATION STUDY ON THE SECOND-CC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
Base	57.73	41.04	29.93	22.68	20.59	48.71	66.04	39.51
Base+CFT	58.16	43.36	31.81	23.68	19.25	50.94	67.63	40.38
Base+CFT+DMA	63.24	49.14	39.81	33.36	24.37	53.12	**80.22**	47.77
DANSA	**65.09**	**51.11**	**41.47**	**34.79**	**24.42**	**54.19**	80.10	**48.38**

TABLE VII
RESULT OF ABLATION STUDY ON LEVIR-CC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Memory Size	Param.	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
500	0.26M	83.07	75.36	69.27	64.48	39.77	73.37	135.83	78.36
1000	0.51M	**85.42**	**77.54**	70.96	65.72	**40.76**	75.56	**139.63**	**80.42**
1500	0.77M	84.45	77.24	**71.03**	**65.83**	40.38	74.51	139.41	80.03
2000	1.02M	78.28	69.92	61.55	52.79	35.07	**77.99**	134.24	75.02
2500	1.28M	79.06	69.55	60.61	51.71	34.44	72.74	140.10	74.75

TABLE VIII
RESULT OF ABLATION STUDY ON WHU-CDC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Memory Size	Param.	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
500	0.26M	80.47	74.09	70.06	67.10	44.23	78.80	146.88	84.25
1000	0.51M	**86.27**	**81.53**	**77.85**	**75.06**	**47.70**	**81.21**	**158.21**	**90.55**
1500	0.77M	82.89	77.77	74.50	72.51	44.59	78.80	148.57	86.12
2000	1.02M	85.97	80.92	76.74	73.51	46.69	80.46	156.43	89.31
2500	1.28M	84.28	79.16	75.94	74.00	45.95	79.23	147.61	86.70

TABLE IX
RESULT OF ABLATION STUDY ON SECOND-CC. THE OPTIMAL RESULTS ARE MARKED IN BOLD

Memory Size	Param.	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	S_m
500	0.26M	61.52	46.36	36.88	30.59	23.13	51.21	74.08	44.75
1000	0.51M	**65.09**	**51.11**	**41.47**	**34.79**	24.42	54.19	80.10	**48.38**
1500	0.77M	64.51	50.98	41.46	34.64	**24.50**	54.04	79.71	48.22
2000	1.02M	61.85	47.33	36.46	27.62	22.20	53.90	**85.19**	47.23
2500	1.28M	62.64	47.99	36.75	28.02	22.74	**54.90**	84.43	47.52

DANSA, it can be seen that aligning the pretrained CLIP change text features by the optimal transport method allows the decoder to better understand the semantic features of the image pairs, which in turn improves the final performance. In summary, by optimizing the representation of difference features and the alignment of visual and text features, fully equipped DANSA demonstrated better overall performance.

2) *DMA With Different Memory Size:* In this section, we perform a study on the number of difference memory banks chosen for the DMA on three datasets. Tables VII–IX list the experimental results, revealing that selecting a memory dimension of 1000 for three datasets demonstrates better performance. For the LEVIR-CC dataset, a memory size of 1000 is sufficient to store the changing target objects and attribute information. For smaller memory sizes, such as 500, model performance shows a slight decline. When using excessively large memory sizes like 1500 and above, learning becomes difficult, preventing the model from learning the changing semantic information, thereby reducing model performance. For the WHU-CDC dataset, its higher resolution means that images of the same size contain simpler scene information but richer object details. Compared to the LEVIR-CC dataset, it is smaller in scale, making the selection of an appropriate memory size critical. Table VIII shows that model performance is constrained when the memory unit is set to 500. Although the target categories are the same as in the LEVIR-CC dataset, the WHU-CDC dataset contains richer intraclass information, also requiring a larger memory size for storage. The DANSA achieves its best performance when the memory size is set to 1000. On the SECOND-CC dataset, it can be seen that a memory size of 1000 is sufficient to address its diverse challenges. A memory size of 1500 also yields good performance, but the number of parameters increases accordingly.

3) *CFT with Different Layer Number:* While introducing the CFT module, the impact of the number of CFT cycles on the generated caption quality is evaluated. Fig. 6 indicates that the model performs optimally when the number of cycles is set to 1. This corresponds to the pseudo-change variations caused by cross-temporal imaging in the dataset. The objective of CFT is to overcome alignment changes between images before and after transformation through attention mechanisms, enabling extracted image semantic

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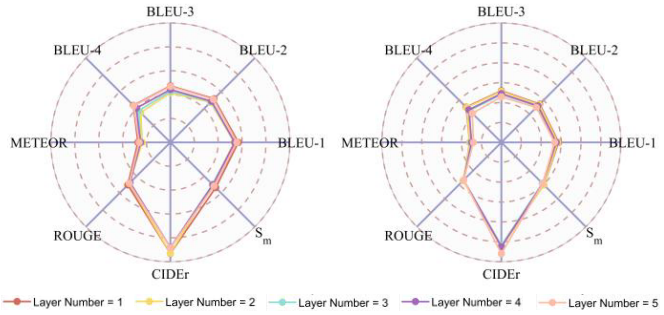


Fig. 6. Comparison of CFT with different layer numbers on the LEVIR-CC and the WHU-CDC datasets.

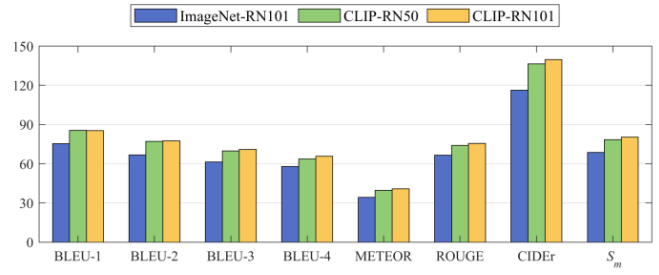


Fig. 7. Comparison results of DANSA with different pretrained backbones on the LEVIR-CC dataset.

features to better align with the change semantic features. Preimage and postimage are primarily influenced by lighting variations, resulting in significant stylistic differences between them. Consequently, after a single CFT interaction, the model achieves satisfactory alignment. However, as interactions increase, the performance on both datasets diminishes to varying degrees. This may weaken the target object and its dependencies within the change.

4) *Comparison of Different Vision-Language Backbones:* To demonstrate the effectiveness and superiority of the CLIP pretrained backbone, we perform experiments on three types of pretrained vision-language features and analyze ablation results. Fig. 7 displays ablation experiments on different backbones. For a fair comparison, the experiments use identical settings to compare features extracted from ResNet-101 pretrained on ImageNet, ResNet-50 pretrained on CLIP, and ResNet-101 pretrained on CLIP across the LEVIR-CC dataset. Experiments show that when the pretrained backbone architecture is the same, different pretraining methods can have a significant impact on the performance of downstream tasks. Compared to features generated using ImageNet, visual-language features generated using CLIP perform better on evaluation metrics. This is closely tied to the simple pretraining method of CLIP and its large-scale image–text dataset. This indicates that CLIP significantly outperforms other pretrained visual backbone models, making it highly valuable for remote sensing image caption generation.

5) *Comparison of Alignment Techniques:* Furthermore, we compare the proposed CSA with the standard global contrast alignment. The comparison results are presented in Fig. 8, where the performance of the model with CSA is improved to different degrees. Compared to the original global contrast alignment, which simply encodes each image or text separately into global features to compute similarity, the proposed CSA can more flexibly adapt to the complex relationships between change semantic features and corresponding text. By jointly optimizing difference feature representation and fine-grained correspondences, CSA captures local details while ensuring global semantic consistency, thus producing more accurate change captions.

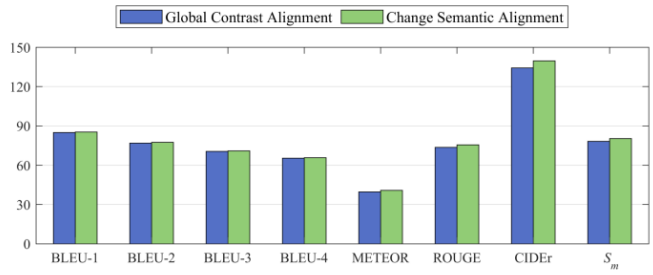


Fig. 8. Comparison results of DANSA with different alignment approaches on the LEVIR-CC dataset.

TABLE X
COMPUTATIONAL COST COMPARISON OF DIFFERENT METHODS IN FLOPS

Method	MModelCC	KGBDNet	D3_Net	DANSA
FLOPs	3.82G	5.64G	43.15G	3.93G

6) *Efficiency and Complexity Analysis:* This section provides a comparison of the computational complexity of the proposed method with state-of-the-art methods, especially the trainable parameters, total parameters, and computational cost in FLOPs. In Figs. 4 and 5, we present plots showing the relationship between the number of trainable parameters in the model, the total number of parameters, and the S_m score. It is evident that DANSA achieves performance comparable to methods requiring full-parameter training while requiring the fewest trainable parameters. Compared to comparison methods that freeze a similar number of parameters, DANSA also achieves impressive results. In addition, in Table X, we compare the FLOPs of three comparative methods with those of DANSA. DANSA achieves a FLOPs value of 3.93 G, representing the lowest computational cost.

G. Qualitative Analysis

1) Change Caption Generation Results: To assess the effectiveness of the change captions generated by the DANSA, we conducted a qualitative evaluation by selecting representative scenes from both the LEVIR-CC, WHU-CDC, and SECOND-CC datasets. As illustrated in Fig. 9, the images on the left depict remote sensing images of the same area captured at different times. The captions on the right represent the reference label, baseline prediction results, and predictions

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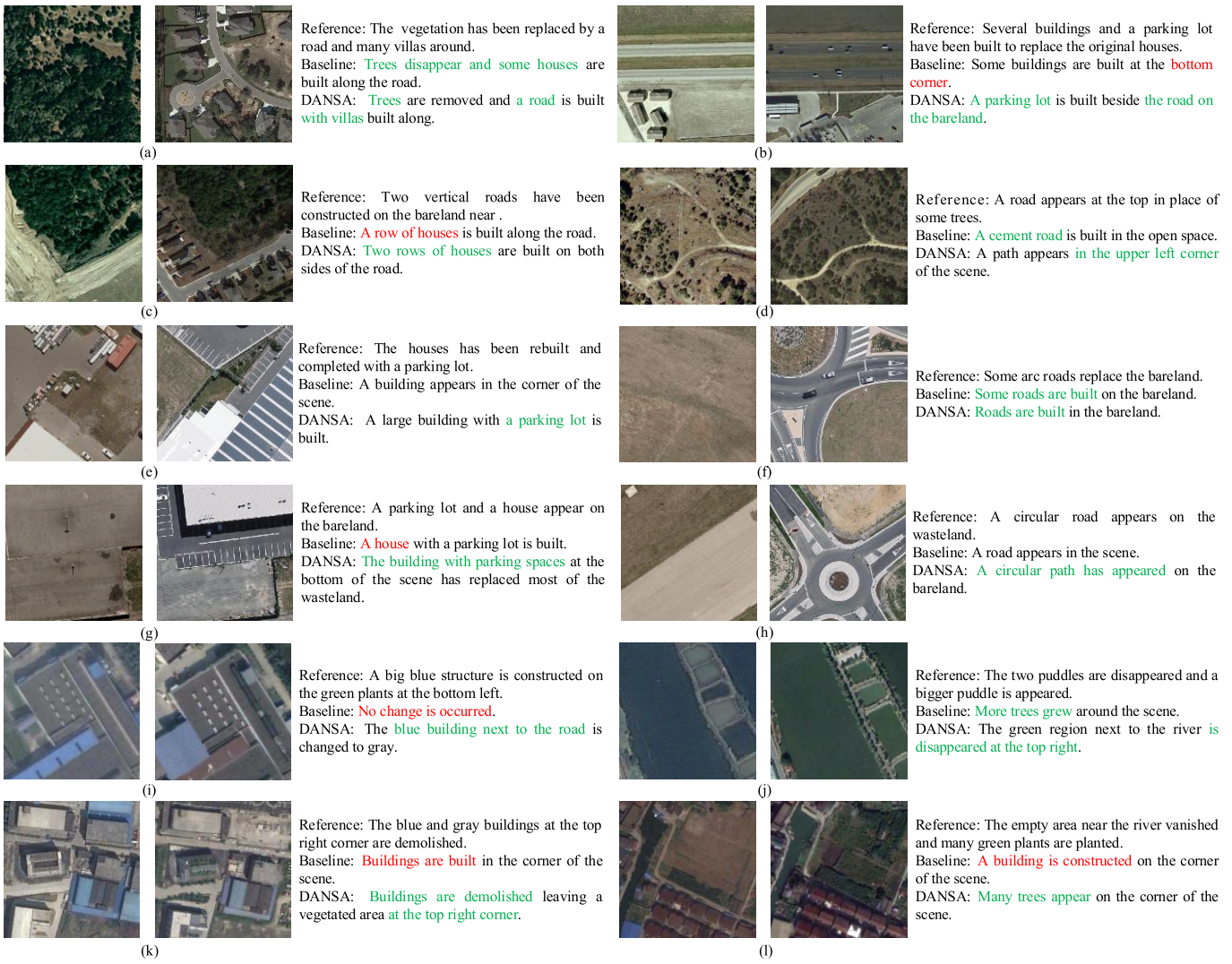


Fig. 9. RSCC results achieved by the proposed method and baseline method on three datasets. Correspond to (a)–(d) LEVIR-CC dataset, (e)–(h) WHU-CDC dataset, and (i)–(l) SECOND-CC dataset. The reference represents ground-truth change captions. Some detailed and accurate words are marked in green. The words that are inconsistent with the image content are marked in red.

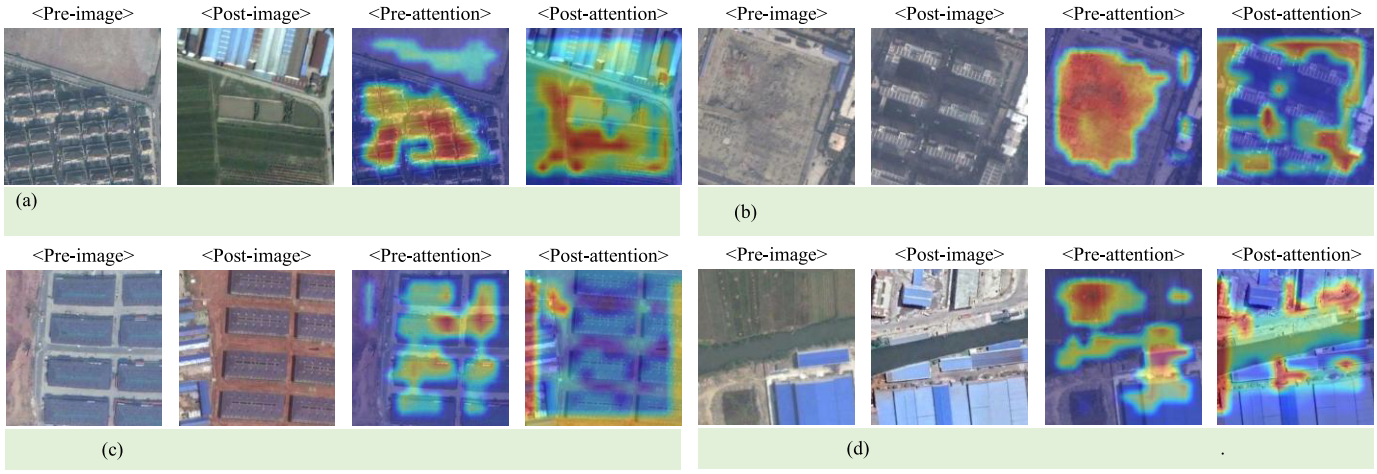


Fig. 10. Visualization of attended image regions along the caption generation process for DANSA on the SECOND-CC dataset. The red color indicates areas with close attention. (a) New rectangular containers were constructed on the opposite side of the road from where the residential area had disappeared. (b) High-rise black houses and asphalt roads are built in place of the empty area. (c) In the empty space at the left edge, blue buildings are built. (d) Around the river, more buildings are constructed.

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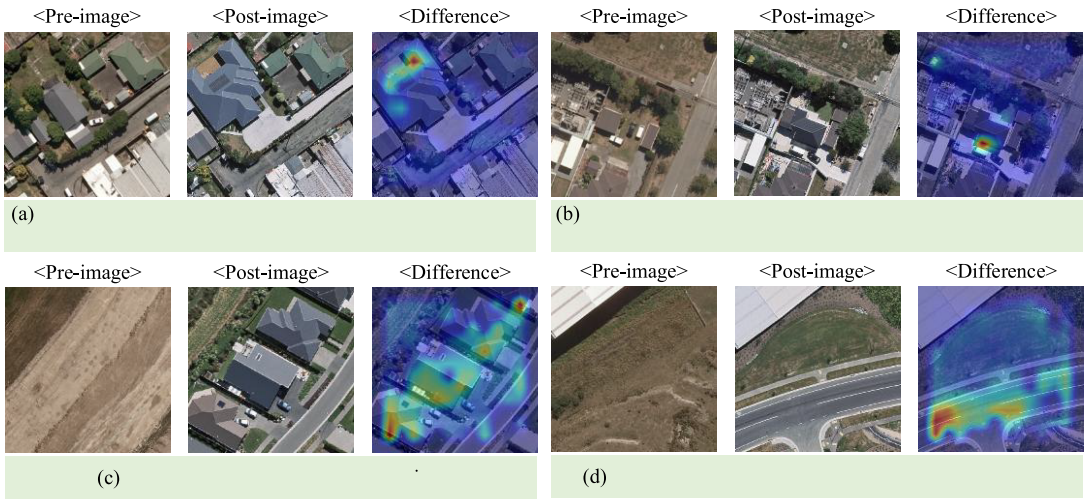


Fig. 11. Visualization of differences in the regions of change in the DANSA on the WHU-CDC dataset. The red color indicates areas with close attention. (a) New house appears in the residential area on the bare land along the roads. (b) Villa has replaced a small house in the original residential area. (c) Several houses appear on the roadside. (d) Crossing roads appear at the bottom of the scene.

generated by our model, respectively. On the LEVIR-CC dataset, the captions generated by DANSA describe the objects more accurately and with a lower error rate than the baseline approach. For example, in the image pair shown in Fig. 9(b), the parking lot is not clearly distinguishable and can easily be confused with the building. In such cases, it is very difficult for the baseline method to generate sentences that include the parking lot, and the results lack interpretability. Furthermore, the captions generated by the baseline method do not include the relative road of the location where the change occurred. However, our model is able to efficiently detect the appearance of the parking lot and reflect it in the generated sentences. In addition, the appropriateness of the sentences meets the requirements. On the WHU-CDC dataset, due to the higher resolution of the images, the objects exhibit richer internal features. It is evident that the proposed method can capture more comprehensive attributes of change objects. For example, the capture of roads with a circular property by the model in Fig. 9(h). This is further illustrated on the SECOND-CC dataset, where the types of changes are more diverse and the scale is smaller, and DANSA is still able to accurately capture these changes. For example, in Fig. 9(k), DANSA can accurately describe the damaged building in the top right corner of the image. All these sentence improvements are attributed to the fact that our difference feature aggregation can query historically relevant features during difference feature learning, thus reducing the interference of false variations. It further enables fine-grained interaction between visual and textual features through a CSA module. Finally, the superiority of the model is also demonstrated through other examples.

2) Visualization of the Attention: To better illustrate the effectiveness of the DANSA, we visualize the feature attention maps of the CFT and DMA models aligned via CSA to demonstrate the role of the proposed module. Fig. 10 shows the feature attention maps of the DANSA for the "before" and "after" images in the SECOND-CC dataset. It can be seen that the activated regions correspond to the areas before and after the change. For example, in Fig. 10(a), the attention is focused on the residential area that has disappeared and the newly constructed rectangular containers. In Fig. 11, we present attention visualizations of the difference features on the postchange images from the WHU-CDC dataset. It can be seen that the model is able to accurately locate the positions of changes, whether in areas with minor local alterations or large-scale changes. For example, the newly constructed house in the upper-left corner of Fig. 11(a) and the newly built road in Fig. 11(d).

TABLE XI
FINE-GRAINED ERROR ANALYSIS OF DIFFERENT METHODS

Error Type	Base	DANSA	Drop	Reduction
Object Omission	41	26	15	36.6%
Wrong Direction	12	0	12	100%
Hallucination	4	2	2	50.0%
Spatial Error	2	0	2	100%
Semantic Contradiction	0	0	0	-

H. In-Depth Quantitative Analysis

1) Fine-Grained Error Analysis: To mitigate the limitations of automatic evaluation metrics, we conduct a fine-grained error analysis on 150 randomly selected test samples from the SECOND-CC dataset. We categorize caption errors into five types: object omission, wrong direction, hallucination, spatial error, and semantic contradiction. Table XI compares the error reduction performance of the proposed DANSA and the baseline. The statistical results reveal that DANSA effectively mitigates most caption errors. Compared with the baseline, DANSA reduces object omission errors to 26 cases, achieving a 36.6% error reduction. This validates the capability of the DMA module to extract global difference features and accurately retrieve changed objects. Notably, DANSA completely eliminates all wrong-direction and spatial errors, reaching a 100% error reduction for both categories. This confirms that our method precisely perceives the directional and spatial variations of scene objects. Moreover, DANSA

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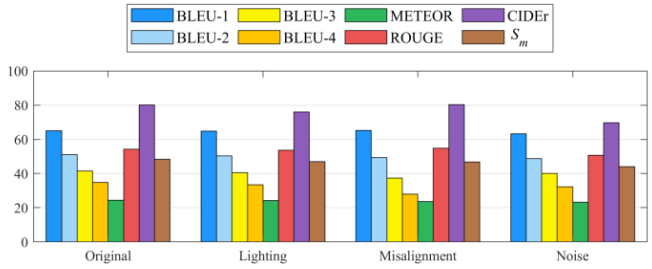


Fig. 12. Comparison results of robustness under controlled nuisances on the SECOND-CC dataset.

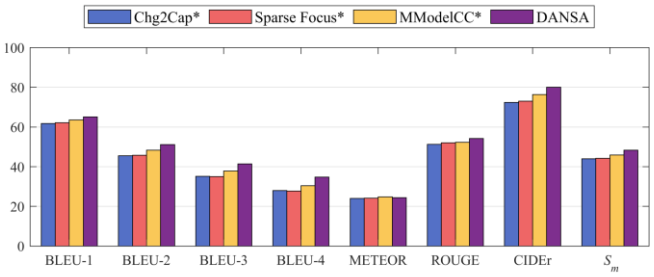


Fig. 13. Comparison results of matched parameter budgets on the SECOND-CC dataset.

reduces hallucination errors by 50.0%, demonstrating that the CSA module provides reliable semantic guidance and suppresses false object generation. No semantic contradiction errors are observed in either method, indicating stable semantic rationality of all generated captions. In summary, the fine-grained analysis verifies that DANSA excels in comprehensive object extraction and accurate change description, which is consistent with the core design goals of its internal modules.

2) Robustness Under Controlled Nuisances: To evaluate the robustness of the DANSA against various common temporal and spatial interference in RSCC scenarios, we conducted comprehensive controlled perturbation analyses based on realistic image degradation. Three typical image disturbance strategies were adopted to simulate complex imaging interference. We randomly adjusted image brightness with a scaling factor ranging from 0.8 to 1.2. We also injected Gaussian noise with sigma values ranging from 5.0 to 10.0 and introduced spatial registration errors by shifting images horizontally and vertically within a range of one to three pixels in both positive and negative directions. All perturbations were applied to the second image of the SECOND-CC test set without model retraining, to quantitatively measure the performance degradation of DANSA and baseline methods, as shown in Fig. 12. The experimental results demonstrate that the proposed DANSA model only suffers slight performance degradation under all four types of interference. Benefiting from the cross-temporal attention design of the CFT module, DANSA stably captures effective change features from distorted and misaligned image pairs.

3) Comparison Under Matched Parameter Budgets: To further enrich comparisons under matched backbone-frozen settings, we constructed backbone-frozen variants for three representative SOTA methods, Chg2Cap [62], Sparse Focus [31], and MModelCC [55], denoted as Chg2Cap*, Sparse Focus*, and MModelCC*. For these three variants, we fully freeze their ResNet-101 visual backbone and only train their downstream encoder, cross-modal interaction, and decoder modules. We conduct quantitative evaluation and intuitive comparison of all four backbone-frozen models on the SECOND-CC dataset, as illustrated in Fig. 13. The experimental results clearly show that freezing the ResNet-101 visual backbone network leads to varying degrees of performance degradation for Chg2Cap*, Sparse Focus*, and MModelCC*. This suggests that the initial performance advantages of these three methods depend largely on fine-tuning the full backbone network to adapt to the features of remote sensing bitemporal change data. In contrast, leveraging the carefully designed CFT, DMA, and CSA modules in the DANSA framework, the proposed model achieves state-of-the-art performance in change captioning generation among all compared methods under the same "frozen backbone network" setting. This fully validates that DANSA exhibits greater robustness and superior adaptability in the context of limited-parameter training for RSCC tasks.

V. Limitation and Future Work

Despite the promising results achieved by DANSA across multiple benchmark datasets, the proposed framework has certain limitations in handling remote sensing images containing small change objects. Specifically, the model sometimes fails to accurately describe minute objects that occupy only a small portion of the image or objects that resemble the background. As shown in Fig. 9(f), although the DANSA can accurately describe the roads in the image, the model struggles to distinguish vehicles traveling on the road. This is likely because their shapes closely resemble those of road traffic signs, and they occupy a similar proportion of the image area. Furthermore, in Fig. 9(j), similar cases such as the disappearance of two puddles are incorrectly described as the disappearance of green areas. The current methods employ single-scale image features extracted by visual encoders to represent objects. Although relatively efficient, this approach lacks a comprehensive multiscale feature representation, resulting in poor capture capability for small-scale change objects in remote sensing images. To tackle this constraint, future research may seek to integrate multiscale feature representations of images into the model framework while balancing computational overhead, thereby improving the recognition sensitivity of the model for small objects. This would better meet the needs of practical applications.

VI. Conclusion

In this work, we propose a novel method named DANSA, which performs RSCC through global difference feature aggregation and CSA. Building upon the pretrained CLIP architecture, this method integrates three modules: CFT, DMA, and CSA. To enhance the difference feature representation capability, we propose CFT to efficiently extract target features across images at different time phases. To fully leverage the relevant information in the historical samples, feature storage

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is performed via DMA, and global relevant difference feature vectors are aggregated through a hard contraction operation to enhance the difference representation of the current change scene. Finally, by cross-modal CSA, the pretrained knowledge of the CLIP text encoder is fully leveraged to achieve fine-grained mapping from vision to language. Comparative analysis with state-of-the-art methods shows that DANSA outperforms other models for change captioning on three benchmark datasets. In addition, the contribution of each module is analyzed through an ablation study.

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